WORKED SOLUTION

STEP II 2017, Question 4

Pure • Cauchy-Schwarz Integral Inequality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q4

The key idea

The Schwarz inequality bounds the integral of a product, $\int fg$, above by the integrals of the squares, $\int f^2$ and $\int g^2$. So the whole skill is choosing f and g so that every integral is one we can compute, except possibly the single one we want to bound. If the integral we want is $\int fg$, we get an *upper* bound. If it is $\int f^2$ or $\int g^2$, dividing through gives a *lower* bound.

Solution

Throughout, write $(*)$ as $\left(\int fg\right)^2 \leq \int f^2 \cdot \int g^2$.

Part (i)

The target has e^t in it, so we take $f(x) = 1$, $g(x) = \mathrm{e}^x$, $a = 0$ and $b = t$. Then

$$\int_0^t \mathrm{e}^x \,\mathrm{d}x = \mathrm{e}^t - 1, \qquad \int_0^t 1 \,\mathrm{d}x = t, \qquad \int_0^t \mathrm{e}^{2x} \,\mathrm{d}x = \tfrac{1}{2}(\mathrm{e}^{2t} - 1),$$

so $(*)$ says

$$(\mathrm{e}^t - 1)^2 \leq \frac{t}{2}\,(\mathrm{e}^{2t} - 1) = \frac{t}{2}\,(\mathrm{e}^t - 1)(\mathrm{e}^t + 1).$$

Since $t > 0$ we have $\mathrm{e}^t - 1 > 0$, and also $\mathrm{e}^t + 1 > 0$. Dividing by the positive quantity $(\mathrm{e}^t - 1)(\mathrm{e}^t + 1)$ keeps the direction of the inequality:

$$\frac{\mathrm{e}^t - 1}{\mathrm{e}^t + 1} \leq \frac{t}{2},$$

as required.

Part (ii)

We want $\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x$ to appear as $\int g^2$, so we take $g(x) = \mathrm{e}^{-\frac{1}{4}x^2}$ with $f(x) = x$ on $[0,1]$. The product $x\,\mathrm{e}^{-\frac{1}{4}x^2}$ is exactly the kind of thing we can integrate (the power of x outside is the derivative of the exponent, up to a constant):

$$\int_0^1 x\,\mathrm{e}^{-\frac{1}{4}x^2}\,\mathrm{d}x = \left[-2\,\mathrm{e}^{-\frac{1}{4}x^2}\right]_0^1 = 2(1-\mathrm{e}^{-\frac{1}{4}}), \qquad \int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}.$$

So $(*)$ gives

$$4(1-\mathrm{e}^{-\frac{1}{4}})^2 \le \frac{1}{3}\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x,$$

and multiplying by 3,

$$\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x \ge 12(1-\mathrm{e}^{-\frac{1}{4}})^2.$$

Part (iii)

Write $J = \int_0^{\frac{1}{2}\pi} \sqrt{\sin x}\,\mathrm{d}x$. All integrals here are over $[0, \frac{1}{2}\pi]$, where $\sin x \ge 0$ and $\cos x \ge 0$, so all the roots are real.

Upper bound. Here J should be $\int fg$. Take $f(x) = 1$ and $g(x) = \sqrt{\sin x}$, so that $g^2 = \sin x$ is easy to integrate:

$$J^2 \le \int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x \cdot \int_0^{\frac{1}{2}\pi} \sin x\,\mathrm{d}x = \frac{\pi}{2}\cdot 1.$$

Since $J > 0$, taking positive square roots gives $J \le \sqrt{\pi/2}$.

Lower bound. Now J must be one of the squared integrals, so we take $f(x) = (\sin x)^{1/4}$, giving $f^2 = \sqrt{\sin x}$.

? **Ask yourself** Which g makes both $\int fg$ and $\int g^2$ computable?

We need g^2 to be easy to integrate, and $(\sin x)^{1/4} g(x)$ to be easy too. The choice $g(x) = \cos x$ does both: $(\sin x)^{1/4}\cos x$ is a power of $\sin x$ times its derivative. The shape of the answer is a clue as well, since a π in the denominator suggests $\int g^2 = \int_0^{\frac{1}{2}\pi} \cos^2 x\,\mathrm{d}x = \frac{\pi}{4}$. Indeed

$$\int_0^{\frac{1}{2}\pi} (\sin x)^{1/4}\cos x\,\mathrm{d}x = \left[\tfrac{4}{5}(\sin x)^{5/4}\right]_0^{\frac{1}{2}\pi} = \frac{4}{5}$$

and

$$\int_0^{\frac{1}{2}\pi} \cos^2 x\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \tfrac{1}{2}(1+\cos 2x)\,\mathrm{d}x = \frac{\pi}{4}.$$

So $(*)$ gives

$$\frac{16}{25} \le J\cdot\frac{\pi}{4}, \qquad \text{that is,} \qquad J \ge \frac{64}{25\pi}.$$

Putting the two halves together,

$$\frac{64}{25\pi} \le \int_0^{\frac{1}{2}\pi} \sqrt{\sin x}\,\mathrm{d}x \le \sqrt{\frac{\pi}{2}}.$$

Aside. Other choices work too, and some do better. Taking $f = (\sin x)^{1/4}$, $g = (\sin x)^{3/4}$ gives $1 \le J \int_0^{\frac{1}{2}\pi} (\sin x)^{3/2}\,\mathrm{d}x$, and a second use of $(*)$ with $f = \sin x$, $g = \sqrt{\sin x}$ shows that $\int_0^{\frac{1}{2}\pi} (\sin x)^{3/2}\,\mathrm{d}x \le \sqrt{\pi/4}$. Together these give the sharper bound $J \ge 2/\sqrt{\pi} \approx 1.128$.

Checks

- (i): as $t \to 0$, the left-hand side is $\tanh(t/2) \approx t/2$, so the bound is tight for small t. For large t the left side tends to 1, while $t/2$ grows without limit.
- (ii): $\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x \approx 0.8556$, while $12(1 - \mathrm{e}^{-1/4})^2 \approx 0.5871$.
- (iii): numerically $J \approx 1.1981$, which sits between $64/(25\pi) \approx 0.8149$ and $\sqrt{\pi/2} \approx 1.2533$. The upper bound is quite close because $\sqrt{\sin x}$ is fairly flat (near 1) over most of the interval, and Schwarz with $f = 1$ is an equality exactly when g is constant.

Common slips

- Getting the direction of the bound wrong. With the unknown integral as $\int fg$, Schwarz can only bound it from above, so a lower bound needs the unknown as $\int f^2$ or $\int g^2$.
- In (i), dividing by $\mathrm{e}^t - 1$ without saying that it is positive.
- Forgetting to take the positive square root, using $J > 0$, in the upper bound of (iii).

Beyond STEP

The Schwarz (Cauchy–Schwarz) inequality says that the inner product $\langle f, g\rangle = \int fg$ satisfies $|\langle f, g\rangle| \le \|f\|\,\|g\|$, with equality exactly when f and g are proportional. It holds in every inner product space, from vectors in $\mathbb{R}^n$ to spaces of functions, and it is what makes "the angle between two functions" meaningful. That idea underlies Fourier series and least-squares approximation. Strang's *Introduction to Linear Algebra* develops inner products and orthogonality, and Abbott's *Understanding Analysis* shows how such inequalities are used to control integrals.

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